

Frédérique LECOURTIER

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PhD student

(Last updated on August 5, 2026)

Research interests

Scientific computing · Physics-informed machine learning · Numerical analysis · Finite element methods · Hybrid solvers · Partial differential equations · Error estimation

Education

- 2023 - 2026 **PhD** in Applied Mathematics, *University of Strasbourg*, France
(in progress) *Team:* Mimesis, Inria Center at the University of Lorraine.
Supervised by: Emmanuel Franck, Michel Duprez and Vanessa Lleras.
- 2021 - 2023 **Master** in Scientific Computing and Mathematics of Information (CSMI), *University of Strasbourg*, France - Honors
- 2018 - 2021 **Bachelor** in Applied Mathematics, *University of Strasbourg*, France - Honors
- 2018 **Scientific Baccalaureate**, *Lycée LeCorbusier*, Illkirch, France

Experience

- 2023 - 2026 **PhD** in Applied Mathematics, *Mimesis*, Strasbourg, France
(in progress)
- Python PyTorch Jax FEniCSx ScimBa
- Subject:* Finite element methods and neural networks for parametric PDEs
Objectives:
- Develop hybrid methods combining finite element methods and neural networks to solve parametric PDEs.
 - Develop scientific machine learning methods in ScimBa (a Python library under development in the Macaron Inria team) for solving parametric PDEs.
- 🔗 [GitHub repository.](#)
- 2023 **Internship**, *Mimesis*, Strasbourg, France
(6 months)
- Python Tensorflow PyTorch FEniCS GitHub VSCode
- Subject:* Finite element methods and neural networks for augmented surgery.
Supervised by: Emmanuel Franck, Michel Duprez and Vanessa Lleras.
- 🔗 [GitHub repository.](#) 🔗 [Report.](#)
- 2022 **Internship**, *Cemosis*, Strasbourg, France
(2 months)
- C++ Python MPI Feel++ Antora Sphinx Doxygen VSCode CMake
- Subject:* Parareal method for solving the heat equation in Feel++.
Supervised by: Christophe Prudhomme and Luca Berti.
Objectives:
- Understand and implement the Parareal method in Python and C++, a parallel-in-time integration method.
 - Implement the heat equation solver in Feel++ (a C++ library developed by the Cemosis team, IRMA).
- 🔗 [GitHub repository.](#) 🔗 [Report.](#)

Publications

- 2026 H. Barucq, M. Duprez, F. Faucher, E. Franck, F. Lecourtier, V. Lleras, V. Michel-Dansac, N. Victorion. **Enriching continuous Lagrange finite element approximation spaces using neural networks**. Mathematical Modelling and Numerical Analysis (M2AN). (pdf), (preprint [arXiv, HAL])

Conferences, presentations and posters

- 2026 **SciCADE 2026**, George Square campus, Edinburgh, United Kingdom
3 July [Website](#). [Slides](#).
- 2026 **CANUM 2026**, Saint-Jacut-de-la-Mer, France
4 June [Website](#). [Slides](#).
- 2025 **PhD seminar**, IMAG Institute, Montpellier, France
17 December [Website](#). [Slides](#).
- 2025 **ICOSAHOM 2025**, McGill University, Montreal, Canada
13-18 July [Website](#). [Slides](#).
- 2025 **EMS-TAG-SciML-25**, Polytechnic University of Milan, Milan, Italy
24-26 March [Website](#). [Poster](#).
- 2025 **DTE & AICOMAS 2025**, Arts et Metiers - ENSAM, Paris, France
17-21 February [Website](#). [Slides](#).
- 2024 **CJC-MA 2024**, Ecole Normale Supérieure de Lyon, Lyon, France
28-30 October [Website](#). [Poster](#).
- 2024 **SciML 2024**, UFR de mathématique et d'informatique, Strasbourg, France
8-12 July [Website](#). [Poster](#).

Teaching experience

- 2025 - 2026 **Teaching assignment**, UFR de mathématique et d'informatique, Strasbourg, France
50 h of computer science labs - L1 (Python) / L2 (Python).
14 h of integrated lectures in computer science - L2 (Python).
- 2025 **Rendez-vous des Jeunes Mathématiciennes et Informaticiennes (RJMI)**, UFR de
(3-day event) *mathématique et d'informatique*, Strasbourg, France
Supervision of research workshops for a group of high school students as part of the RJMI event.
- 2024 - 2025 **Teaching assignment**, UFR de mathématique et d'informatique, Strasbourg, France
64 h of computer science labs - L1 (Python) / L2 (Python) / L3 (C++).
- 2023 - 2024 **Teaching assignment**, UFR de mathématique et d'informatique, Strasbourg, France
50 h of computer science labs - L2 (Python) / L3 (C++).

Skills

GitHub · Python · C++ · C · Rust · LaTeX · PyTorch · JAX · ParaView · Gmsh · FEniCS

References

PhD advisor:

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PhD co-supervisors:

Michel Duprez : michel.duprez@inria.fr

Vanessa Lleras : vanessa.lleras@umontpellier.fr

Collaborator:

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